

01 - Databases in the Real World

CSCI 112 - Contemporary Databases

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Learning Objectives

- Understand design considerations when choosing a database
- Understand workload requirements for running a database

1

Scenario for the Day



Context Setting

- *ArchiTECH Solutions* just signed a new project with *The Coco Group*, a huge conglomerate in the Philippines. This project will kickoff *The Coco Group's* digital transformation journey. But given its size, they will be starting with their restaurant chain, "*Kahit Saan*" to control the scale of the project.
- *Kahit Saan* would like to have a centralized POS system that is cloud-based, and would have the simple functionality of selling different products in a restaurant setting.

A person with short dark hair, wearing a grey and black striped sweater, is seen from the back, looking at a wall covered in various papers, diagrams, and photos. The papers include flowcharts, sketches, and text. A large blue arrow graphic points from the left towards the top of the wall.

**You are ArchiTECH Solutions' top consultant
when it comes to database technology
architecture.**

How should you approach this requirement?

2

The Well Architected Framework



Well-Architected Framework

- This framework was developed by AWS and uses general concepts that are not specific to the platform. Thus, it can be adopted regardless of the cloud provider you are using. It can even be used for on-prem architectures!
- The well architected framework is anchored by six different pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.

2.1

Operational Excellence



Operational Excellence

Is my application performing as expected (or even better)?

- Operational excellence refers to the ability of your system to support required workloads effectively.
- For databases, this includes choosing the correct database and deployment design in terms of performance and availability.



Performance Considerations

What do we need to consider when choosing a database in terms of performance?

- Required latency - the amount of time a request travels from server to the database and back.
- IOPS (Input Output Per Second) - the performance of storage devices like HDDs or SSDs to process requests. (# of transactions)
- Read/Write throughput - the amount of data that can be read or written in a database system for a specific amount of time. (MB read/written)
- Concurrency - the capability of a database system to process requests at the same time.



IOPS vs Throughput

What is the difference between IOPS and throughput?

- IOPS describes the amount of input and output transactions per second while throughput describes the amount of data transferred per second.
- Consider Shoppee's warehouse:
 - ▷ IOPS describes the number of parcels shipped in and out daily
 - ▷ Throughput describes the weight/peso value of parcels shipped in and out daily
- IOPS and throughput are measured per second.



Availability Considerations

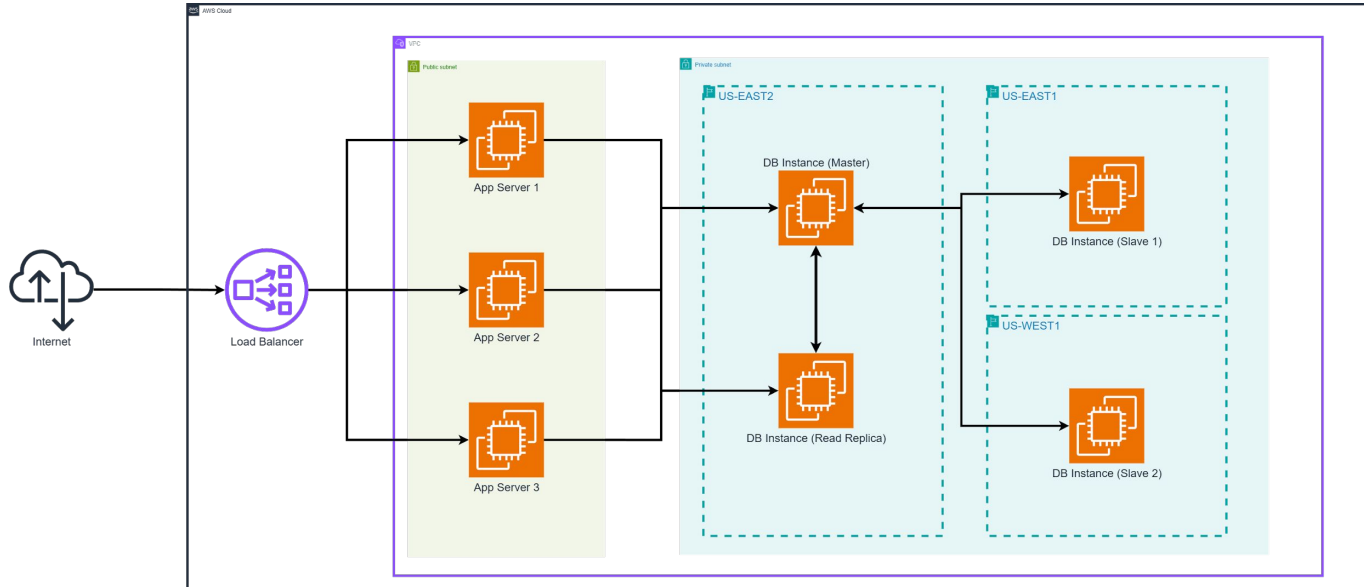
What do we need to consider when choosing a database in terms of availability?

- Read replicas - servers that are dedicated to read operations
- Clustering - decoupled servers (usually in a master-slave configuration) that are used to efficiently distribute database load.
- Geo-distributed deployments - Similar to clustering, but the servers are located in separate physical locations.



Distributed Application Architecture

When one server is not enough...





Choosing the Right Database Type (SQL vs NoSQL – DocumentDB)

- SQL
 - ▷ OLTP
 - ▷ OLAP (at times)
- NoSQL – DocumentDB
 - ▷ High Performance Computing
 - ▷ IoT
 - ▷ Logging
 - ▷ Session State
 - ▷ Messaging

2.2

Security



Security

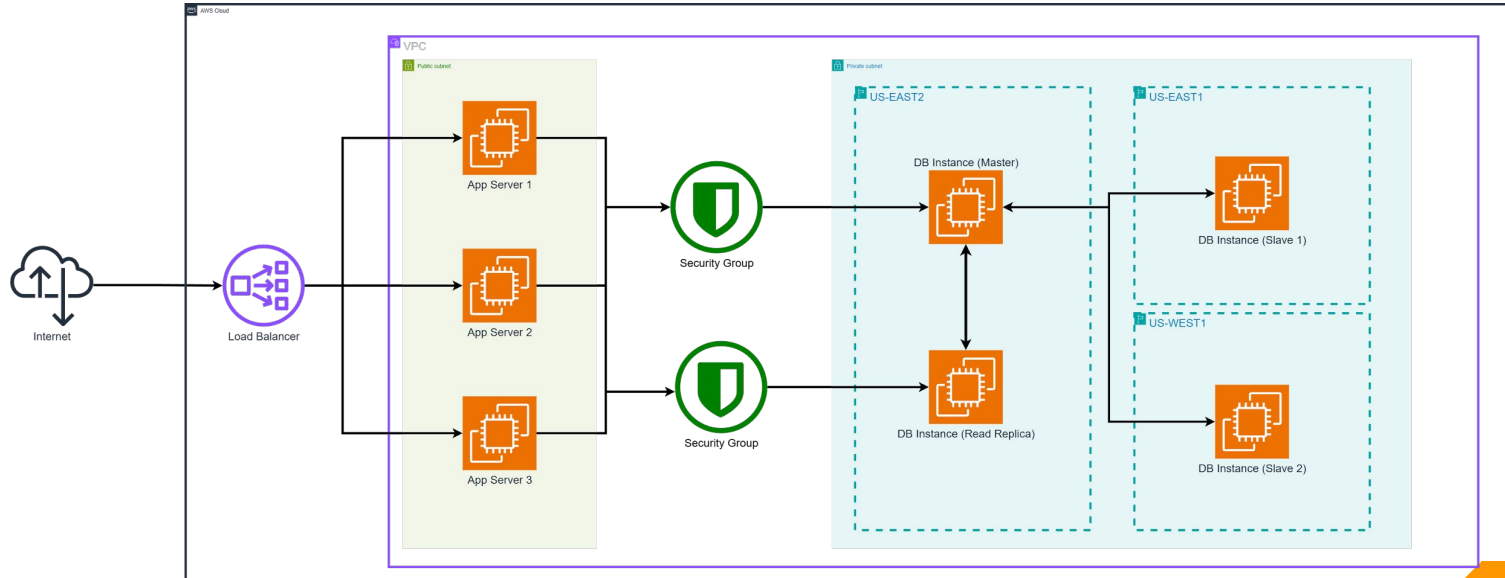
Is my database secured from exploits?

- Security focuses on fortifying all fronts of your application, especially the database since it contains your (sometimes other's) data.
- Security may also refer to proper data protection through cryptography.



Network Architecture for Security

Generic architecture for a secured database





Principle of Least Privilege

- Only give access to whoever need it.
- Explicitly deny access to those who shouldn't have it.

2.3

Reliability

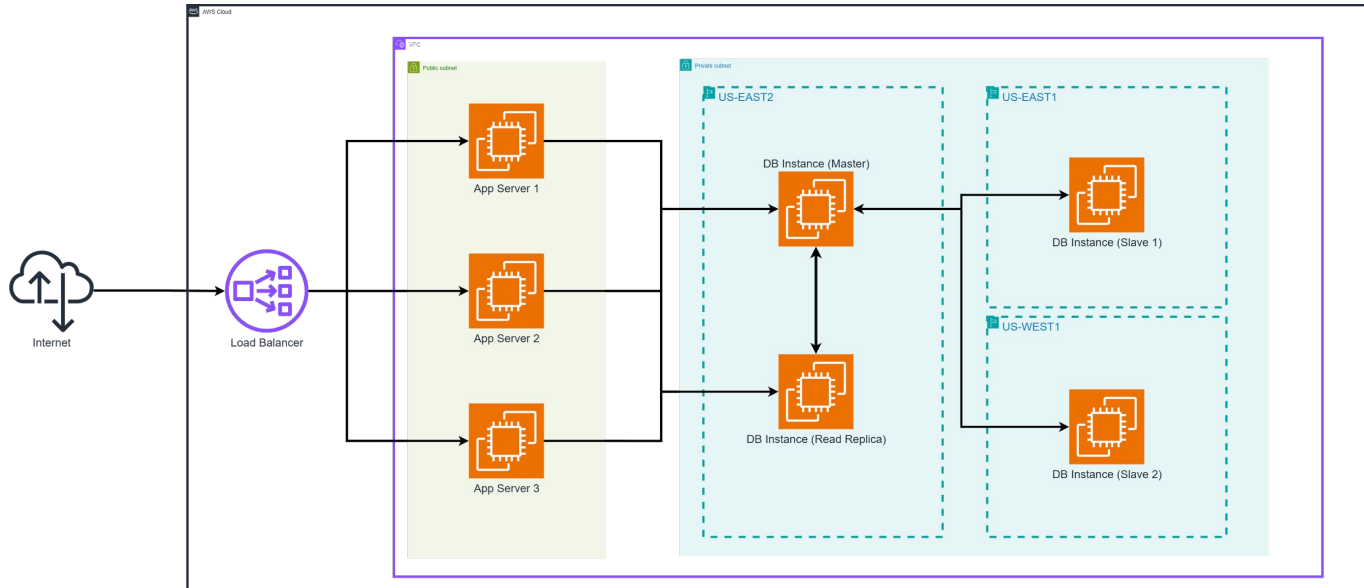


Are You Automating Enough?

- Reliability refers to the ability of your distributed system architecture to address failures automatically, with minimal human intervention.
 - Self-healing - when a server goes down, does it have the ability to switch traffic to an available server while it spawns a new server to replace the faulty one?
 - Auto-scaling - when demand for my applications rise, does my distributed system architecture have the ability to scale and meet the demand?



Reliability





Serverless Architecture

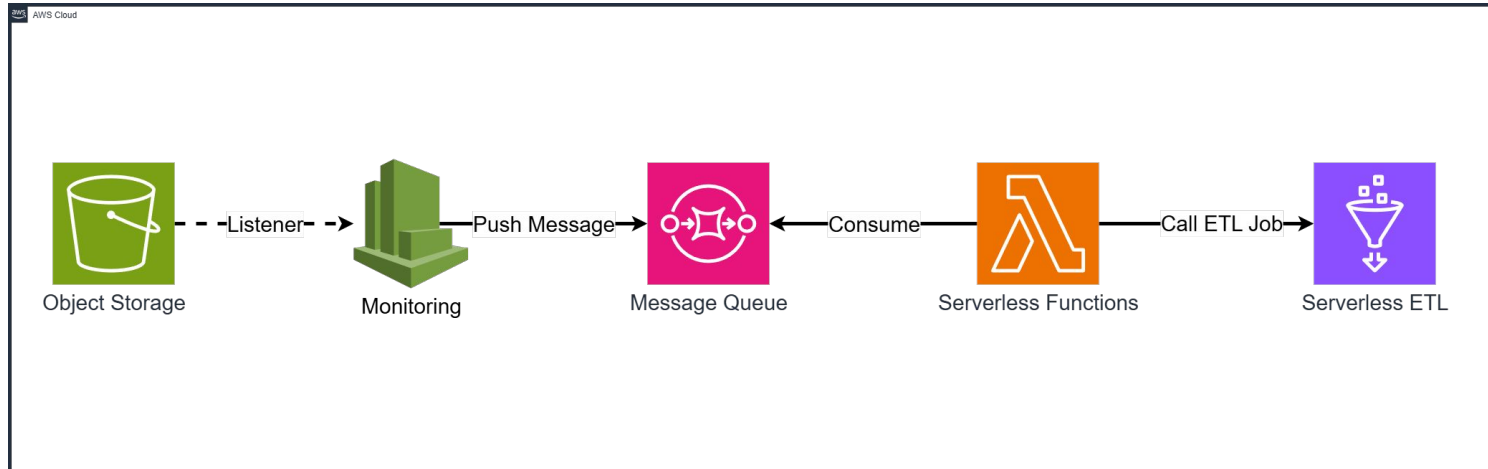
Deploy and Forget

- Serverless architecture refers to the usage of serverless services to satisfy the requirements of your application.
- It does not mean that there is no server. The server is abstracted from the developers who only need to deal with their code and the application architecture.



Serverless Architecture

Example Use Case





Benefits of Serverless

- High availability - the cloud provider will be the one to allocate physical resources to meet your application's demand
- No server maintenance - since the resources are abstracted, there is no need to worry about the underlying servers
- Automatic replications and backups - the cloud provider also takes care of the replications and backups for your databases, which is helpful when your database needs to scale to meet demands.

2.4

Performance Efficiency



Performance Efficiency

Is your architecture efficient?

- Performance efficiency refers to the ability to use computing resources and databases efficiently to meet system requirements as they evolve.
- Best practices:
 - ▷ Conducting design and architecture reviews
 - ▷ Decommissioning of unused resources
 - ▷ Resource utilization tracking
 - ▷ Using the right services/platforms for the right use cases



Operational Efficiency for Databases

Am I using the correct database for the correct workload?

- File System
- Object Store
- Relational Databases
- Non-relational Databases



The Vs of Big Data

Am I handling my data efficiently using my chosen database platform?

- Volume - How big the data is
- Velocity - How fast the data is moving
- Variety - In what formats do the data come in

2.5

Cost Optimization



Cost Optimization

How can you optimize your architecture to spend less money?

- Cost optimization refers to the ability to run your systems and delivery your business value with the lowest cost possible.
- Best practices:
 - ▷ Implement financial management in your resources
 - ▷ Automated budget tracking
 - ▷ Resource optimization for under utilized resources

2.6

Sustainability



Microsoft Submerged Data Center



3

Group Work



Groupings

Group 1

Name, Name, Name, Name

Group 2

Name, Name, Name, Name

Group 3

Evangelista, Manibog, Navarro,
Paderayon, Zapico

Group 4

Name, Name, Name, Name

Group 5

Name, Name, Name, Name



Requirement

As a database technology architect, I want to create a cloud-based POS architecture for *Kahit Saan* so that they can track their sales in their restaurants.



Deliverables

- Choose a database type (SQL vs Document DB) - 30 points
 - Indicate the considerations and decision points in choosing the database type
 - How would you describe the data's "Vs"?
- Through a diagram, show which components communicate with the others (especially the database) - 10 points
- How are you securing your database? - 10 points
- The use of any concept is allowed, even outside the boundaries of this lecture.

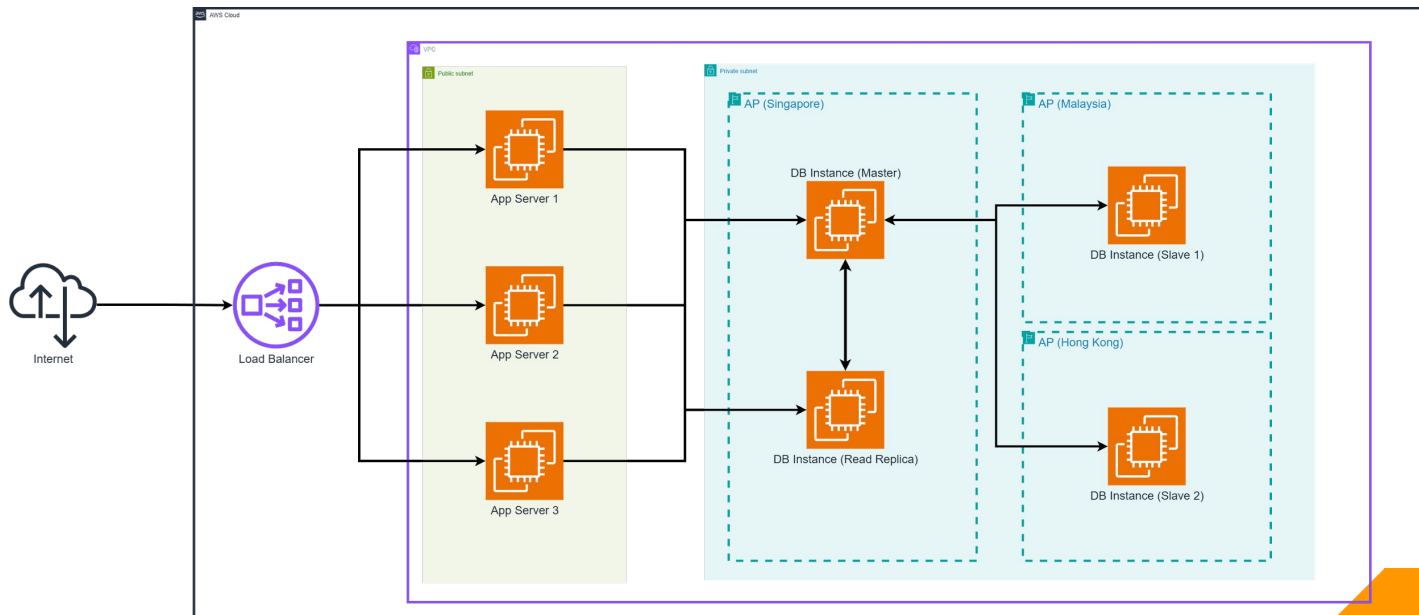
3.1

My Take



Distributed Version

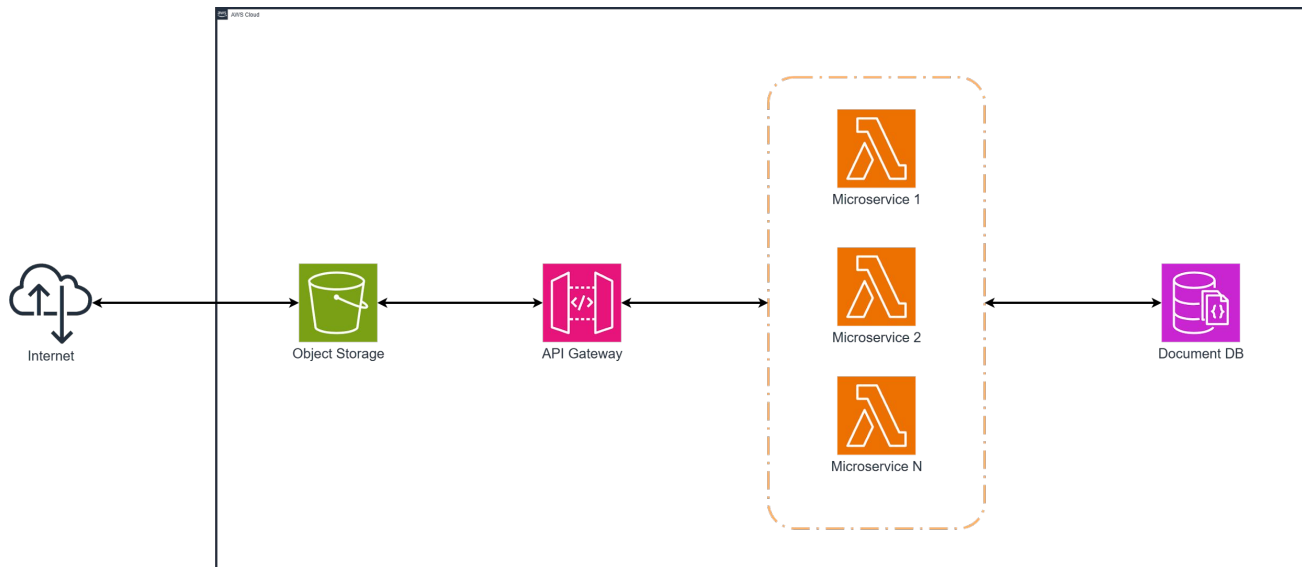
Using both managed and non-managed components





Serverless Version

Only using serverless components



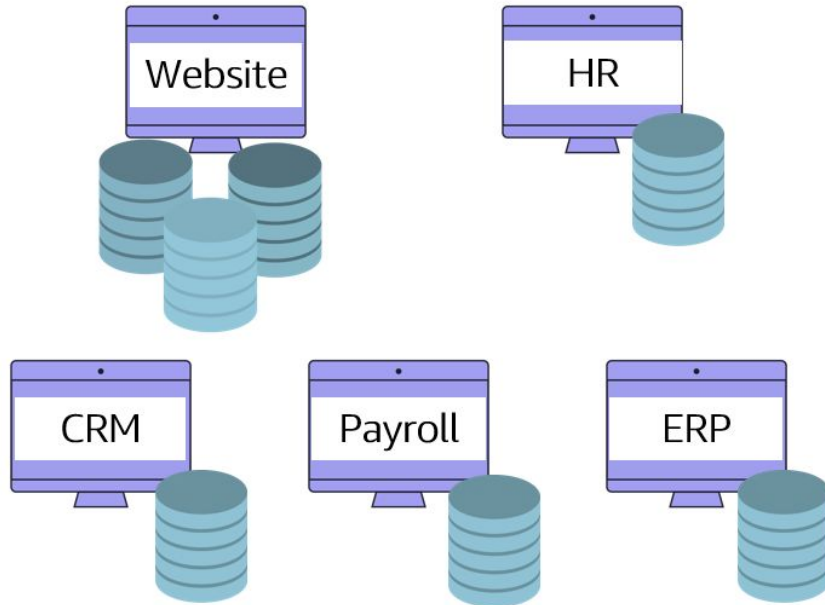
A person with short dark hair, wearing a grey and black striped sweater, is seen from behind, looking at a wall covered in various design sketches, photos, and notes. A large white box with the word "Questions?" is superimposed over the person's head. The background is a collage of creative work, including wireframes, photos of people, and handwritten notes. A blue arrow points from the left towards the top of the image.

Questions?

3.2

Comparison of DB Systems

Characteristics of traditional applications



Users	100s to 1,000s
Data volume	GB to TB
Locality	HQ
Performance	Seconds
Request rate	Tens of thousands
Access	Internal servers, PC
Scale	Up
Economics	Pay upfront
Developer access	Days/weeks/months

Characteristics of traditional applications



Users	One million plus
Data volume	TB to PB to EB
Locality	Global
Performance	Milliseconds to microseconds
Request rate	Millions
Access	Mobile, IoT, devices
Scale	Up-down, out-in
Economics	Pay as you go
Developer access	Instant API access

End.

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